

Their platform supports drones that don't necessarily need 5G, drones that don't have high speed data transmission requirements but just want bandwidth for command and control operations over a large area, Dodla suggested. "We provide long range Wi-Fi beyond visual line of sight which lets you connect to a drone that's more than 20 km away, which you can't do in existing Wi-Fi technologies," he said, as beaming Wi-Fi signals over long distances is detrimental to the already limited power capacities of tiny drones. This is an optimised way of staying connected to and commanding drones that are quite a long distance away, a solution that quite a few of Qualcomm's drone partners already use.

"With the success of the Aero India in Bengaluru, a lot of drone companies want to deploy ad-hoc networks – which essentially means deploying a private 5G network in an area where you can't deploy a 5G network, because there are no towers. So you start hovering these drones in the air that act as 5G transmitters and receivers to create a bubble – whether across a few mountains – which can be quickly brought down after completion of work," Dodla highlighted.

This is an indication of 5G and drone use not just on the client front but also as a network node, giving rise to interesting use cases of these technologies. Imagine a swarm of drones forming a mesh network, either for an entertainment company or the military, "Not necessarily on the receive side but also on the transmission side," he emphasised.

Rural 5G coverage

Not so much yet in India, specifically, but Qualcomm is an uptick in 5G mmWave deployments in rural sectors. "Much to our surprise, there are deployments of 5G mmWave, for example, in rural Americas and rural Australia. You typically think mmWave can only do 100-150 metres max in terms of range. Yes, that may be the case in dense buildings and urban centres, but in the rural areas where you don't have a lot of these obstructions, we've been able

to see kilometres worth of range for 5G mmWave," Dodla mentioned.

The gains are huge, he further said, because for a fraction of the cost, it enables broadband access to rural communities, according to Dodla. "Currently, if you look at our penetration map around the world, the dense urban areas are very well covered by all these operators. But it's the rural areas which are very sparse, right, and, and they're relying on satellite service or no service. But now operators can go with millimetre wave solutions, where a single mmWave transmitter with directional antennas can cover probably 20-30 homes. It instantly brings good broadband coverage, which can potentially improve their economy with access to online services."

Whether it's rural 5G deployments or public private partnerships, Dodla told me how Qualcomm is engaging with the Indian government. "With regards to the issue of safe and reliable devices in India's 5G infrastructure, we have provided a list of best practices (to the Indian government). We've had several sessions with key officials in MeitY and the Department of Telecom, where we talked about current vulnerabilities, where do they exist, where are they typically found, in what applications, how are backdoors exploited, what are some of the instances where backdoors have been exploited? When it comes to critical infrastructure deployments, we want to make sure that the decision makers in the government know about these things and how to navigate them," emphasised Dodla.

A lot of this work was done throughout 2020 in the middle of the pandemic, suggested Dodla, as they put together a white paper to educate key officials in the right government ministries which actually regulate the telecom-related sector.

AI on device

Qualcomm's very well known for its modem technology, but not as well known for some of the other stuff that

we do very well, according to Dodla – which is AI on device. When everyone and their old grandmother is talking about AI, he mentioned how AI is something that Qualcomm's been doing for a long time.

"We've had AI on our camera platforms, in our robotics platforms, and we've spoken in the past how a lot of this AI stuff can't be cloud native, because it's too expensive, and that's why on-device generative AI is a cornerstone of our product strategy," he emphasised. Qualcomm has invested a lot of effort into this area, which we're going to see in products launching later this year – instances of Stable Diffusion and generative AI use cases that aren't cloud native but device native. "We're looking forward to seeing that, as AI's going to be a big part of what we're going to do in the future," said Dodla. It's all about enhancing existing product lines, as on-device generative AI solutions Qualcomm's building starts going into people's hands.

In terms of challenges to 5G and AI roll-out, the only pain point is actually managing demand, believes Uday Dodla. "As an ecosystem, we have to figure out how to showcase the advantages of automation using connected technologies, a prime example of which is smart electricity metres. These connected metres have high availability and reliability requirements, which is supported by our cellular technology. The only thing we have to crack is to create the demand for these devices, by talking more about their benefits both for consumers as well as power companies," he said.

The second thing Dodla highlighted is to work with our partners in the operator space to ensure access technologies don't get fragmented. "We have too many choices right now in the form of LPWA, LoRaWAN, RF Mesh, and more. And that brings me to my third concern, where we are seeing these non-standardized solutions getting deployed, which is something of a challenge that we need to overcome as an industry going forward," Dodla emphasised. ■